

ARYA Y P

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Experience

- Samsung Prism – Intern** (GitHub) Feb 2026 – July 2026
- Developed a Node.js/Express + Socket.io signaling server for a decentralized calling app, handling room orchestration, WebSocket-based peer discovery, and offer/answer/ICE message relay across nearby devices.
 - Architected the Android client for real-time audio/video call aggregation using Wi-Fi Direct P2P networking, hardware-accelerated H.264 encoding, a multi-stream MCU mixing architecture, and STUN/TURN/ICE NAT traversal with fault-tolerant host-failover and automatic peer re-election.
- Tech Stack:** Node.js, Express, Socket.io, WebRTC, Wi-Fi Direct, H.264, STUN/TURN/ICE

Education

- B.E. in Information Science** 2023–Present
BMS College of Engineering, Bengaluru
CGPA: 9.69
- Rashtrottana Vidya Kendra, Bengaluru** 2021–2023
Percentage: 94
- Vidya Vikasa High School, Chitradurga** 2020–2021
Percentage: 98.4

Projects

- Resilience Grid – AI-Driven Energy Supply Chain Resilience for Import-Dependent Economies** (GitHub)
- Designed an operational intelligence platform for India's strategic import corridors, synthesizing GDELT geopolitical events, OFAC sanctions, AIS vessel tracking, and commodity feeds into composite corridor risk scores refreshed every 10 minutes with WebSocket-pushed alerts.
 - Implemented a geospatial digital twin and scenario simulation engine, including a PuLP linear-programming SPR drawdown optimizer with Monte Carlo p10/p50/p90 uncertainty bands and a 63-cell stress-test matrix.
- Tech Stack:** Python, FastAPI, Pydantic, PuLP, Gemini AI, React, Vite, Tailwind CSS, Zustand, Recharts, Leaflet, WebSockets
- KSP Crime Intelligence – Bilingual FIR Database Assistant** (GitHub)
- Created a deployed bilingual (English/Kannada) NL-to-SQL assistant over a 30-table normalized FIR schema, with role-based access control enforced at the SQL layer, criminal-network visualization, and 30-day predictive forecasting.
 - Delivered voice interaction, context-aware multi-turn conversations, geospatial hotspot mapping, and full audit logging with server-side PDF export and honest service degradation across a fallback pipeline.
- Tech Stack:** Python, FastAPI, SQLite, Anthropic Claude, Tailwind CSS, Leaflet, Chart.js, Zoho Catalyst
- Esell – Multi-Currency Resell Marketplace with Multi-Agent RAG Assistant** (GitHub)
- Developed a full-stack resell marketplace (Node.js, Express, Socket.io) with multi-currency live-FX pricing, locked-rate Stripe checkout, cart payments, real-time chat, and offer negotiation.
 - Deployed a multi-agent RAG assistant (FastAPI, LangGraph, Qdrant, Gemini) with query routing, a Self-RAG self-correcting retrieval loop, and guardrails for semantic product search and grounded product Q&A.
- Tech Stack:** Node.js, Express, Socket.io, Python, FastAPI, Google Gemini API, Render

Skills

- Languages:** C, C++, Python, Java
- Frameworks & Libraries:** React, Node.js, Express, FastAPI, Socket.io, Tailwind CSS, Vite, TensorFlow, Scikit-learn, OpenCV, HuggingFace Transformers
- Databases:** MySQL, MongoDB, PostgreSQL, SQLite, Redis, Supabase, Qdrant
- Machine Learning:** DL, NLP, ANN, RNN, CNN, RAG, Computer Vision
- Networking & Real-Time:** WebRTC, Wi-Fi Direct, STUN/TURN/ICE, WebSockets

Publications

- Network Guardian: An Intelligent System That Automatically Protects Computer Networks from Attacks Using Artificial Intelligence** – ICWITE 2026

Certifications

- Full Stack Web Development – Udemy
- Data Science, ML, DL, NLP Bootcamp – Udemy
- Google Cloud Engineering Certificate – Google Skills

Achievements and Activities

- HyperAPI Global Hackathon – Winner.**
Built an end-to-end pipeline for processing financial documents and detecting anomalies scoring 97.5% using HyperAPI.